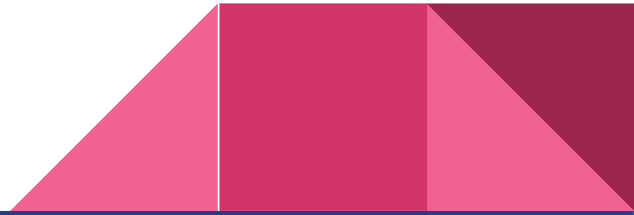
Project INSPIRE Meeting III

Liam Gallagher

Agenda

- Groups Made
- Leads Decided
- Digital Tasks
- Analog Example Design
- Announcement



Groups

- Analog and Digital Groups decided, list will be posted after the meeting!

- These are not restrictions on what you learn but will be what you are tasked with providing.

Leads

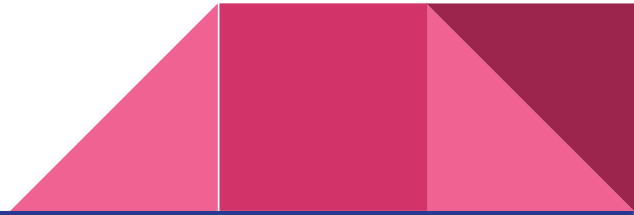
- Analog Lead:
  Alex Peacock

- Digital Leads:
  Emilio C. Ruiz
  Tal Avital

- I’m always appreciative of help, so feel free to pitch in and help, your leads (and I) will appreciate it

Part Selection

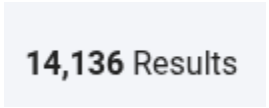
- All projects require part selection to meet requirements
- Datasheet comparison reveals specifications for desired components
- Modern distributor websites enable searching by parameter



Part Example

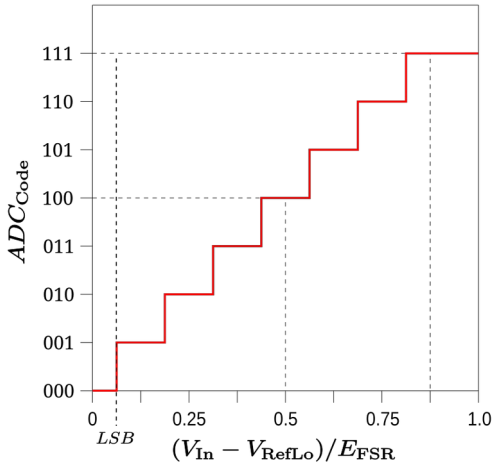
- ADC for Digital Backend is responsible for converting our analog signal to digital byte representation.

- Digikey offers over 14,000 ADCs, no one can sort through them all by hand. We must narrow down the options through parameter filters



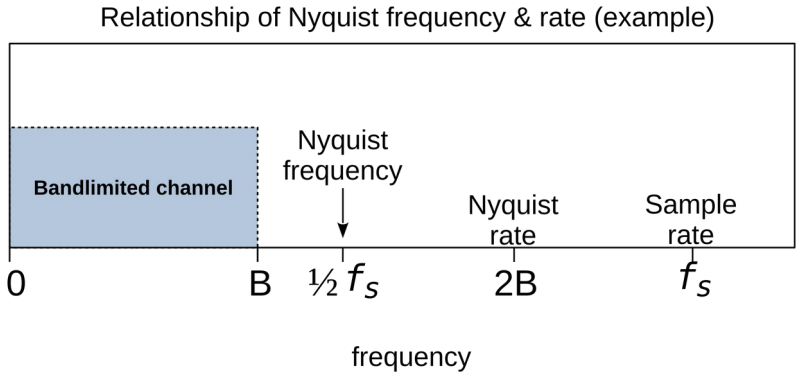
ADC Parameters

- Some of the most crucial parameters for us are:
  - Sampling Frequency: how fast the ADC can make data acquisitions
  - Bit Depth: how small a voltage change can be detected
  - Number of Channels: how many streams can be digitized



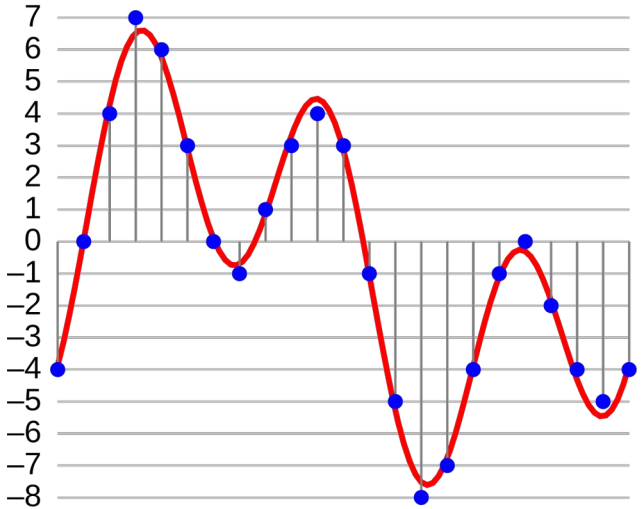
ADC Parameters II

- Sampling Frequency is the highest priority
  - Harry Nyquist discovered any sampling system cannot accurately capture any signal $\geq \frac{1}{2} \cdot f_s$
  - This means in order to capture a signal of frequency f, we must sample at least 2f times a second.
  - 30kHz in our sampling band means we need at least 60kHz



ADC Parameters III

- Bit depth is priority #2
  - Our Signal-to-Noise (SNR) determines how easy we can pick up signals
  - SNR formula for a number of bits is: $SQNR \approx (1.76 + 6.02b)dB$
  - This means, a 12-bit ADC manages ~72dB of range, a 16-bit ADC manages 96dB, etc.
  - In reality, Effective Number of Bits(ENOB) is our actual limit.



ADC Parameters IV

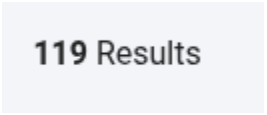
- Required parameters are:

Parameter	Required Value
# of channels	2
Sampling Frequency	⩾96kHz
Input Type	Differential
ENOB	Higher the better (⩾12 bit)

Our Search Continues

- When we account for these filters (and specify audio ADCs), the number of ADCs tightens
- Now, we can filter for things like Active/Not Discontinued, Footprint (Physical Design), and this brings tens of thousands of parts down to a few we can see on one page.

- https://tinyurl.com/yvcdacnr

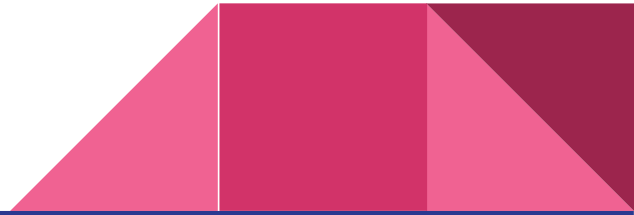


We did it!

- We selected a part! Now, we have to repeat this for every single component in the design, both analog and digital, all with differing parameters.
- As such, the plan is to assign each people to a sub-circuit such that one person isn’t responsible for part selection alone.

Analog vs Digital Part Selection

- Analog design has the advantage of SPICE, where you can download models of the analog component and simulate it to see accurate performance of what the component will behave.
- With LTSpice, changing out a part is as simple as adjusting parameters



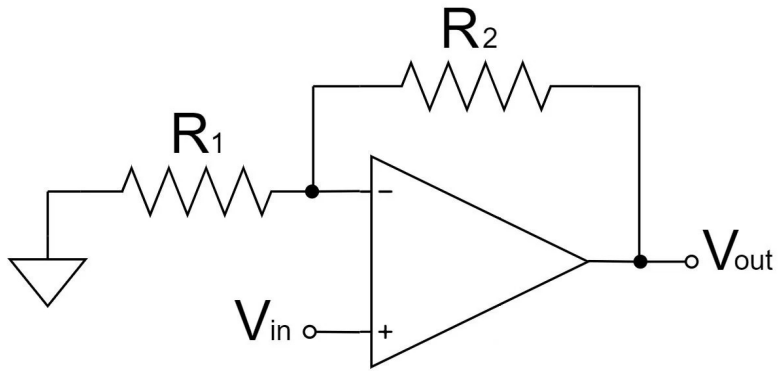
Amplifier Design

- A basic idealized op-amp non-inverting amplifier has a transfer function of:

$$\mathrm{K} = \mathrm{V}_{out}/V_{in} = 1 + \frac{R_2}{R_1}$$

- This assumes a perfect op-amp with an infinite input impedance, and a infinitely fast voltage output

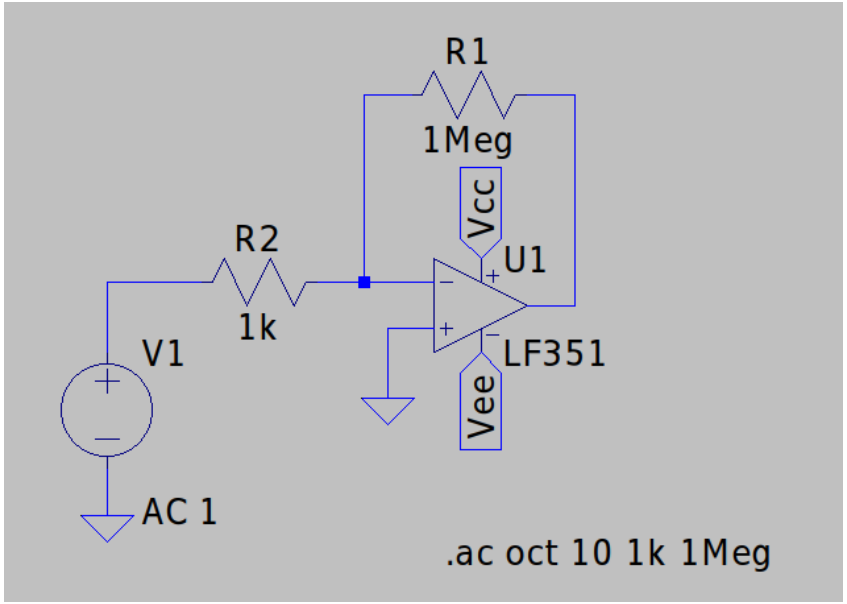
Non-Inverting Amplifier Circuit



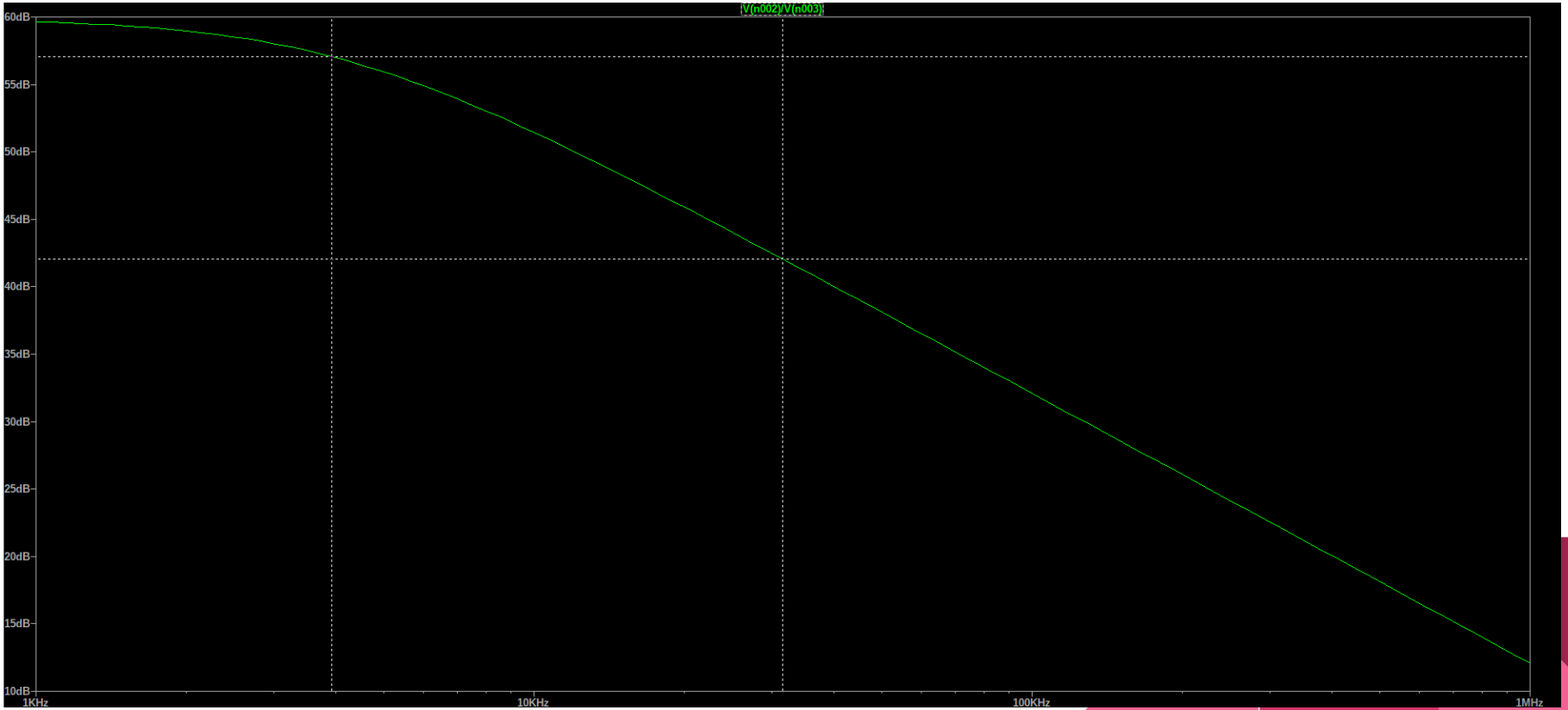
Op-Amp Limitations

- Real Op-Amps have input “leakage” current, and frequency dependent gain that affect the behavior over the spectrum (see next slide)

- This simulation uses a realistic model of a op-amp, which accounts for these effects.



Op-Amp Limitations



Op-Amp Limitations

- 15dB drop in gain means the low frequency content is 5.62x stronger then the high frequency in the passband, which is unacceptable. Also notice how the gain is vastly different then the 1000x we should expect.

- As such, we cannot use this op-amp, or at least not in the gain configuration we want.

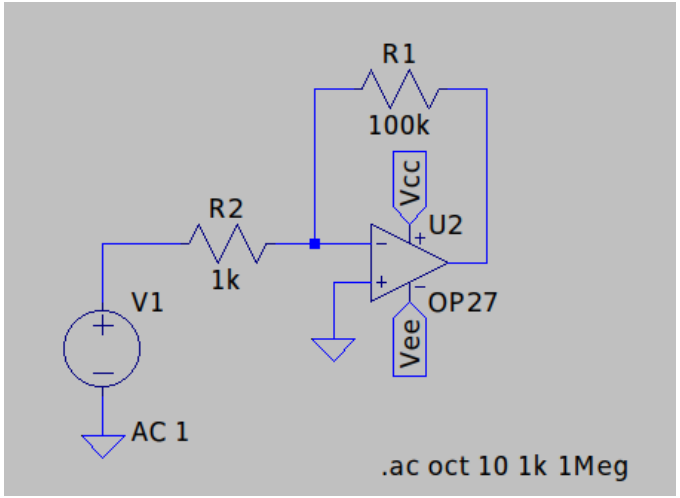
Op-Amp Parameters

- Four major parameters:
  - Open-loop Gain (A)
  - Gain-Bandwidth Product (GBW)
  - Slew Rate (SR)
  - Input Impedance (R_{in})

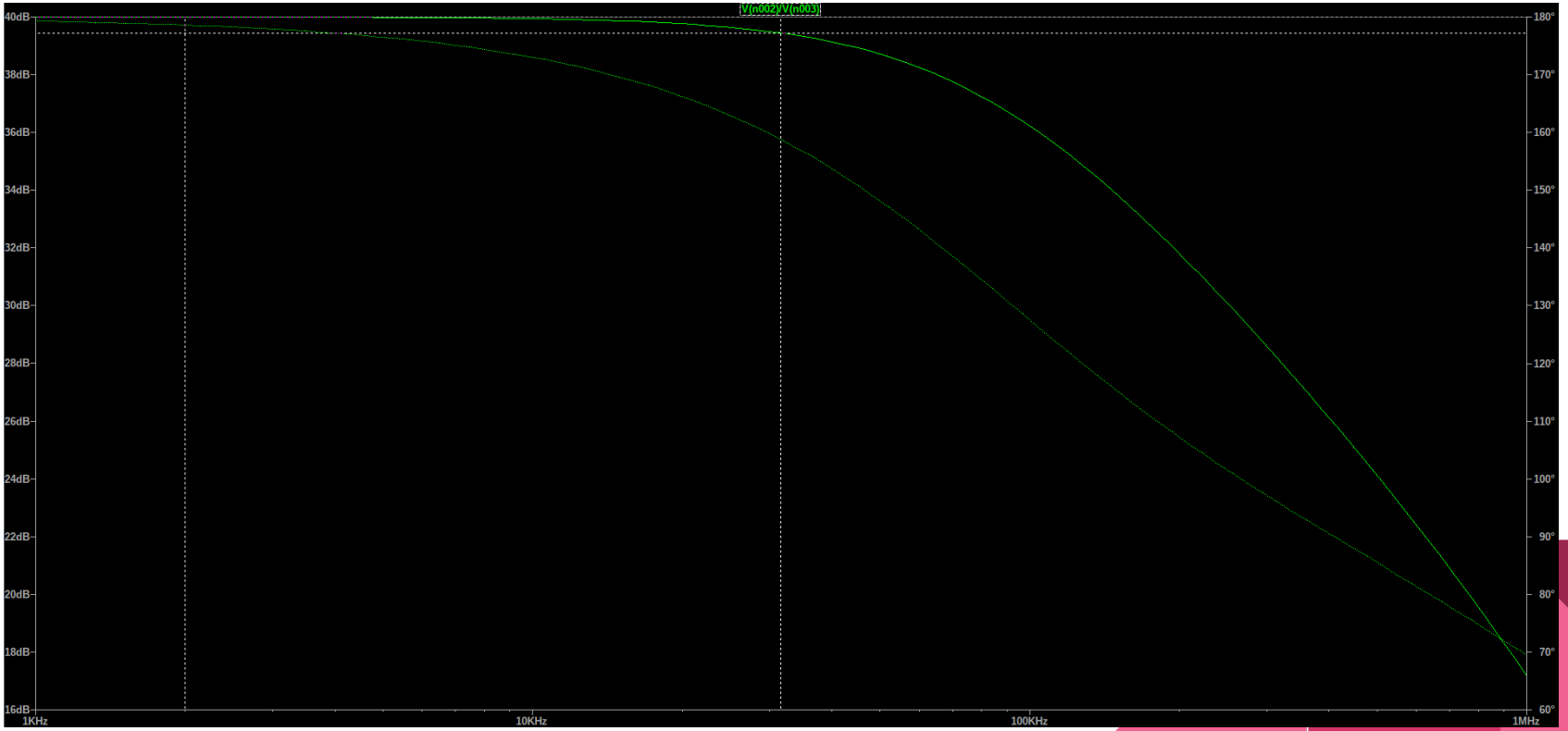


Op-Amp Improvement

- Picking a more modern op-amp with a higher GBW means that we get a higher frequency cutoff for the same gain.
- Restricting gain for each op-amp (building amplifiers in cascade), also means that the individual load on each amplifier is reduced.

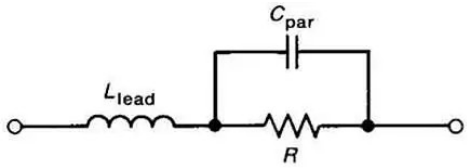


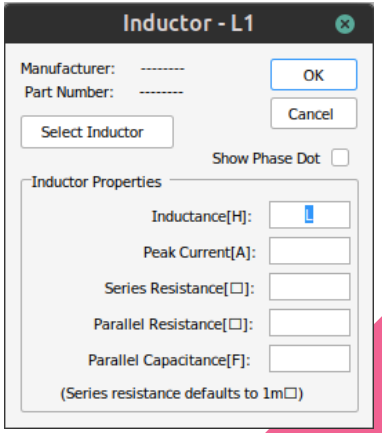
Op-Amp Improvement

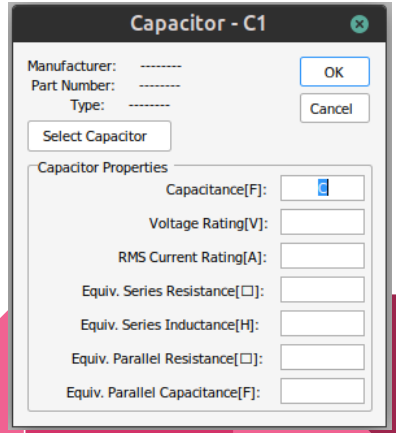


Non-Ideal Components

- Passives (Resistors, Inductors, Capacitors) all have “parasitics” which are effectively extra components in the circuit.
  - This effectively makes each component a passive filter, bad for our observation band.
- These effect the behavior of our amplifiers and filters a bit
- We model them by the series and parallel resistance, inductance and capacitance.

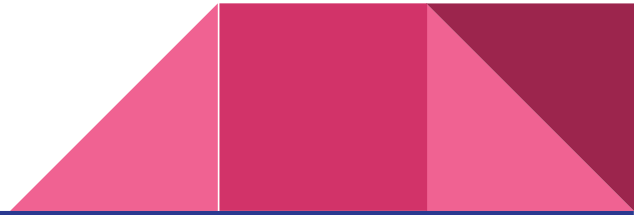






Inserting Circuit Models

- Companies and individuals will make custom SPICE models of parts, which can be imported
- “.include {path}” will add it to the circuit and let you define custom parts
- This lets us import op-amps, transistors and other components that LTSpice does not ship with.



Going Forward

- We will be dividing the team into groups, which will be where you get a sub-circuit (amplifier, filter, ADC, differential converter, etc), which we will provide a short design document with explanation and requirements, and it will be up to you to find components, design the circuit, and make a SPICE simulation.

- Not a weeklong task (do not worry, we are here to help! This is not a grade)